

Beyond Unitary Designs: Quantifying Trainability in Parameterized Quantum Circuits

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Parameterized quantum circuits (PQCs) represent a central class of controllable quantum evolutions, yet their trainability can be severely limited by exponentially vanishing gradients, known as the barren plateau phenomenon. Existing analyses typically rely on strong randomness assumptions, such as unitary designs and remain limited in addressing general structured circuits. In this Letter, we develop a measure-based framework to characterize the trainability of general PQCs and establish a level classification of gradient variance across orthogonal subspaces. Our results reveal a trainability advantage in previously unexplored unitary 1-designs compared to higher-order designs. We further introduce the improvement potential to quantify maximal variance enhancement through adjacent insertion of parametric oracles. This framework provides unified analytical guidance for understanding and engineering trainable quantum circuits on near-term quantum hardware.

Introduction.— Quantum technology has gradually come to play an indispensable role in a wide range of frontier fields such as metrology [1–4], quantum sensing [5–7], and quantum machine learning [8–12]. In the current Noisy Intermediate-Scale Quantum (NISQ) era, hardware imperfections and limited coherence times hinder complex quantum computations, making Variational Quantum Algorithms (VQAs) the dominant paradigm for achieving quantum advantage [13–19]. VQAs constitute a hybrid quantum-classical framework that solves complex problems by optimizing trainable PQCs via classical optimizers (e.g., gradient descent [20, 21]). Therefore, the successful execution of VQAs is highly dependent on the trainability of the chosen PQC. The trainability of a PQC is typically directly governed by the gradient variance of the cost function. When the gradient variance of the cost function vanishes exponentially with the system size, the gradient concentrates around zero, known as the barren plateau [22, 23]. It means that large-scale circuits with this property often require an exponentially large number of measurements to converge, leading to severe trainability issues. Thus, characterizing the factors influencing trainability is crucial for the systematic design of scalable and trainable PQCs.

To investigate the factors influencing the trainability of PQCs, the unitary 2-design framework is commonly used to analyze the gradient variance of a given PQC at a specific parameter [24–30]. The unitary 2-design framework is essentially a simulation of the Haar random distribution, relying on the assumption that the analyzed PQC approximates the Haar measure up to the second moment. However, PQCs that approximate the Haar measure exhibit strong randomness [27], leading to poor trainability and limited practical utility.

In practice, quantum circuits are typically designed with tailored structures for specific problems, which inevitably leads to a lack of randomness. Common examples include shallow

or locality-constrained hardware-efficient ansatzes [31], circuits generated by continuous-time evolution under structured Hamiltonians [32], and symmetry-constrained structures [33]. Such non-random PQCs may exhibit strong trainability and play an important role in practical applications. However, existing unitary 2-design frameworks are inherently tied to the strong randomness assumption; they are effectively restricted to analyzing untrainable circuits, leaving these practical, structured (and potentially trainable) circuits outside their scope. Many studies have attempted to extend the current analytical frameworks. For example, Refs. [34–37] extended the unitary 2-design framework to analyze PQCs composed of multiple unitary 2-design blocks, but each block remains constrained by the strong randomness assumption. The work in Ref. [38] sought to develop a more general analytical approach based on Jordan algebra decomposition. However, applying such frameworks generally requires identifying the underlying mathematical structure of the circuit in advance, which can be highly nontrivial in practice. To systematically analyze the trainability of practical PQCs, it is necessary to propose a more comprehensive gradient variance framework applicable to a broader class of PQCs, and within which the mathematical structure behind a given circuit can be obtained straightforwardly.

In this Letter, we develop a measure-based framework to characterize the trainability of PQCs beyond the standard unitary design paradigm. This framework approximates the distribution of any PQC to arbitrary precision by developing a class of polynomial probability measures, and further establishes explicit mathematical structures for practical circuits. Based on this framework, we establish a level classification for trainability by projecting the gradient variance into different orthogonal subspaces. In particular, we demonstrate the significant advantages of the previously overlooked unitary 1-

designs in terms of trainability. Furthermore, we introduce the concept of “*improvement potential*” to quantify the maximum achievable enhancement of the gradient variance at a given parameter through adjacent insertion of parametric oracles. Our work provides systematic analytical guidance for identifying and enhancing the trainability of parameterized quantum circuits in the NISQ era.

Preliminaries.— PQCs play a central role in near-term quantum computing, serving as the primary framework for implementing variational quantum algorithms. Structurally, a circuit $U(\theta)$ consists of a sequence of parameterized quantum gates, whose tunable parameters θ control the circuit’s evolution. Formally, under the hardware-efficient ansatz, $U(\theta)$ is given by $U(\theta_1, \dots, \theta_L) = \prod_{l=1}^L U_l(\theta_l)W_l$, where $U_l(\theta_l) = \exp(-iH_l\theta_l)$ is a rotation gate and W_l is the entangling gate chosen from a fixed set of unitary operators. The cost function is of the form $C(\theta) = \text{Tr}[U(\theta)\rho U^\dagger(\theta)O]$. Here, ρ is the input density matrix, O is the target observable, and the rotation angles $\{\theta_j\}$ are typically assumed to be independent. Following the standard research on barren plateaus [22, 27], we characterize trainability by the gradient variance.

To calculate the variance, we start by analyzing the structure of the partial derivative $\partial_\ell C := \partial C / \partial \theta_\ell$. Specifically, we split the circuit at the parameter θ_ℓ into two parts, U_L and U_R , i.e., $U = U_L U_R$, where $U_L(\theta) = \prod_{j=\ell+1}^L U_j(\theta_j)W_j$, $U_R(\theta) = \prod_{i=1}^\ell U_i(\theta_i)W_i$. Consequently, the partial derivative $\partial_\ell C$ can be expressed as

$$\partial_\ell C = i \text{Tr}[U_R \rho U_R^\dagger [H_\ell, U_L^\dagger O U_L]]. \quad (1)$$

Let $X_L = i[H_\ell, U_L^\dagger O U_L]$, then the gradient variance can be expressed explicitly as

$$\text{Var}^{(\mu_L, \mu_R)}(\rho, O) = \int d\mu_L(U_L) d\mu_R(U_R) \text{Tr}[U_R \rho U_R^\dagger X_L]^2, \quad (2)$$

where $U_L \sim \mu_L$ and $U_R \sim \mu_R$, which means for any measurable set $A \subseteq \mathcal{U}(d)$, $\mathbb{P}(U_L \in A) = \mu_L(A)$, $\mathbb{P}(U_R \in A) = \mu_R(A)$. Here $\mathcal{U}(d)$ denotes the unitary group of degree d , and $\mathbb{P}$ is the probability distribution function.

Since the form of $\partial_\ell C$ is associated with the 1-moment of U_R and U_L , this dependence motivates the use of unitary designs in the evaluation of gradient variances. Formally, a probability measure μ is said to form a unitary t -design [39, 40] if, for every polynomial $p \in \text{Hom}(\mathcal{U}(d), t, t)$, we have $\int d\mu(U) p(U) = \int d\mu_H(U) p(U)$, where $\text{Hom}(\mathcal{U}(d), t, t)$ denotes the set of all homogeneous polynomials of degree t in both the matrix entries of U and $U^\dagger$, and μ_H is the normalized Haar measure.

As discussed in the Introduction, the assumption of a local unitary 2-design is not suitable for many practical PQCs. A new framework is required to develop a general theory of trainability. In the following, we introduce the framework of polynomial measures, which serves as the foundation for our work.

Polynomial approximation to general PQCs.— When a PQC is equipped with a specific parameter distribution, it induces a probability measure μ . Here we introduce the notion

of a *polynomial measure*. A finite measure μ is said to be polynomial if it is absolutely continuous with respect to μ_H and its Radon–Nikodym derivative $d\mu/d\mu_H$ [41] is a polynomial. The space of polynomial probability measures is dense (in an appropriate sense) in the space of probability measures [42]. In other words, for an arbitrary PQC with a given parameter distribution, there exists a sequence of polynomial probability measures that approximates the corresponding measure to arbitrary accuracy. One can refer to the Supplementary Materials, Sec. III for the rigorous proof. By continuity of the function $(\mu_L, \mu_R) \mapsto \text{Var}^{(\mu_L, \mu_R)}(\rho, O)$, if two pairs of probability measures (μ_L, μ_R) and (μ'_L, μ'_R) produce close expectation values for the relevant observables, the corresponding gradient variances $\text{Var}_\theta^{(\mu_L, \mu_R)}(\rho, O)$ and $\text{Var}_\theta^{(\mu'_L, \mu'_R)}(\rho, O)$ are also close. The proof of continuity would also be provided in the Supplementary Materials, Sec. III. This compatibility implies that the trainability of arbitrary PQCs can be characterized by integrals of polynomial functions with respect to the Haar measure, whose evaluation is enabled by Weingarten calculus.

Given a polynomial probability measure μ with the Radon–Nikodym derivative f , not all monomials in f contribute to the gradient variance. By the Weingarten calculus, only those monomials lying in $\text{Hom}(\mathcal{U}(d), t, t)$ for some $t \in \mathbb{Z}_+$ contribute to the integral [43]. We refer to the sum of such monomials as the *balanced component* of f . The balanced component admits a decomposition in terms of zonal polynomials [44–46], which naturally arise in matrix-variate distributions (e.g., the Wishart distribution) and in integration over compact Lie groups [46]. For our case on the unitary group $\mathcal{U}(d)$, these polynomials are indexed by irreducible representations of $\mathcal{U}(d)$, which are labeled by non-increasing integer sequences $\lambda = (\lambda_1, \dots, \lambda_d)$ with $\lambda_1 \geq \dots \geq \lambda_d$ [47]. Let χ_λ denote the character of the irreducible representation V_λ and d_λ be its dimension. For each such index λ , one can associate a family of zonal polynomials [48], which are defined by

$$Z_{\lambda, V}(U) := d_\lambda \chi_\lambda(V^\dagger U). \quad (3)$$

According to the orthogonality relation $\langle Z_{\lambda, V_1}, Z_{\lambda', V_2} \rangle = \delta_{\lambda, \lambda'} \langle Z_{\lambda, V_1}, Z_{\lambda, V_2} \rangle$, one can expand the balanced component in the basis of zonal polynomials, which yields a decomposition of the gradient variance into independent contributions. Moreover, since $\text{Tr}[U \rho U^\dagger X_L]^2$ is a balanced polynomial in U of degree at most 2, only indices in the set $S_\alpha = \{\lambda : |\lambda| = 0, |\lambda_+| = \alpha\} (\alpha \leq 2)$ contribute to the gradient variance, where $|\lambda_+|$ denotes the sum of the positive entries of λ . There are $\sum_{\alpha=0}^2 |S_\alpha| = 6$ such indices, and the corresponding zonal polynomials are given in [48] explicitly. Further details on the properties of zonal polynomials and their application to gradient variance calculations are provided in Supplementary Materials, Sec. III.

Assume further that the parameters θ_l are i.i.d. and uniformly distributed on $[0, 2\pi)$. For such PQCs, when evaluating the gradient variance using the Weingarten integral, the probability density term can be replaced by the corresponding

zonal polynomial, yielding consistent results. The details are provided in the following proposition.

Proposition 1. Let $U(\theta) = \prod_{l=1}^L U_l(\theta_l) W_l$ be a parameterized quantum circuit composed of fixed entangling gates $\{W_l\}$ and parameter-dependent rotation gates $\{U_l(\theta_l) := \exp(-iP_l\theta_l)\}$, where each P_l is a Pauli word and the parameters θ_l are i.i.d. uniformly distributed on $[0, 2\pi)$. For a Pauli word P , define

$$P_1 := I, \quad P_2 := P, \quad P_3 := \frac{I + iP}{\sqrt{2}}, \quad P_4 := \frac{P + iI}{\sqrt{2}}. \quad (4)$$

Then the following statements hold:

1. **Parameter-dependent rotation gate.** The probability distribution of a parameter-dependent rotation gate $U_l(\theta_l)$ corresponds to the polynomial

$$1 + \frac{1}{2} \sum_{k=1}^2 Z_{(1,0,\dots,0,-1), P_{l,k}} + \frac{1}{4} \sum_{\lambda \in S_2} \sum_{k=1}^4 Z_{\lambda, P_{l,k}}. \quad (5)$$

2. **Fixed entangling gate.** The probability distribution of a fixed entangling gate W_l corresponds to the Dirac distribution concentrated at W_l .
3. **Full circuit.** The probability distribution of the full circuit $U(\theta)$ corresponds to the polynomial

$$1 + \frac{1}{2^L} \sum_{i_1, \dots, i_L=1}^2 Z_{(1,0,\dots,0,-1), \prod_{s=1}^L P_{s,i_s} W_s} + \frac{1}{4^L} \sum_{\lambda \in S_2} \sum_{i_1, \dots, i_L=1}^4 Z_{\lambda, \prod_{s=1}^L P_{s,i_s} W_s}. \quad (6)$$

Proposition 1 provides an explicit polynomial measure family to approximate any PQC under the hardware-efficient ansatz. Thus, the gradient variance of an arbitrary PQC can be obtained by the Weingarten calculus of polynomials with respect to the Haar measure. For rigorous proofs of Proposition 1, one can refer to the Supplementary Materials, Sec. III.

Level classification of variances.— The above analysis shows that an arbitrary PQC can be approximated by polynomial probability measures. The balanced component of such measures, which governs the gradient variance, admits a decomposition in terms of zonal polynomials [44–46]. Consequently, the total variance can be expressed as the sum of three contributions:

$$\text{Var}^{(\mu_L, \mu_R)}(\rho, O) = \sum_{\alpha=0}^2 \text{Var}_\alpha^{(\mu_L, \mu_R)}(\rho, O), \quad (7)$$

where the α -th order variance is defined by

$$\text{Var}_\alpha^{(\mu_L, \mu_R)}(\rho, O) := \sum_{\lambda \in S_\alpha} \int d\mu_L(U_L) d\mu_R(U_R) \left(\int d\mu_H(V) \overline{Z_{\lambda, U_R}(V)} \text{Tr}[V \rho V^\dagger X_\lambda] \right)^2. \quad (8)$$

Through the zonal polynomial decomposition, we have projected the variance onto several subspaces that admit explicit computation successfully. Among these, the bound $\text{Var}_0 = \frac{2d^2}{(d^2-1)^2} \|\tilde{\rho}\|_2^2 \|\tilde{O}\|_2^2$ for the zeroth-order variance is a well-established result [22], where $\tilde{A} := A - d^{-1} \text{Tr}[A]I$ for $A \in \{\rho, O\}$. For the first- and second-order variances, we provide the following theoretical upper bounds.

Theorem 1. Let U_L and U_R be the unitaries obtained from the splitting defined before, with $U_L \sim \mu_L$ and $U_R \sim \mu_R$. Then the α -th-order variance admits the following asymptotic bounds:

$$\text{Var}_\alpha^{(\mu_L, \mu_R)}(\rho, O) \lesssim d^{\alpha-2} \|\tilde{\rho}\|_2^2 \|\tilde{O}\|_2^2 \quad \text{for } \alpha = 0, 1, 2, \quad (9)$$

where $\lesssim$ hides a universal constant.

Building on the uniform approximation by polynomial measures, Theorem 1 characterizes the variance contributions from orthogonal subspaces $\{\oplus_{\lambda \in S_\alpha} V_\lambda : \alpha = 0, 1, 2\}$ for general PQCs, thereby providing a level classification of total variances based on structural properties of measures. It demonstrates that the second-order variance determines whether there exists a barren plateau at the given parameter. Moreover, for a pure state ρ , we can further build a relation between the total variance and the first-order variance by

$$\text{Var}^{(\mu_L, \mu_R)}(\rho, O) \leq \frac{d}{2} \text{Var}_1^{(\mu_L, \mu_R)}(\rho, O) + d^{-1} \|\tilde{\rho}\|_2^2 \|\tilde{O}\|_2^2, \quad (10)$$

indicating that a large first-order variance is necessary to avoid degraded trainability. Readers can refer to the Supplementary Materials, Sec. IV for detailed proofs of Theorem 1 and Eq. (10).

By Eq. (8), the first-order contribution to the variance vanishes if at least one of μ_L or μ_R forms a unitary 1-design. It then follows from Eq. (10) that the total variance is bounded by $O(d^{-1})$. The following proposition establishes that this bound is tight.

Proposition 2. Fix a pure state ρ and an idempotent operator P on $\mathbb{C}^d$ (i.e., $P^2 = I$). For $m \in \mathbb{N}$, define a probability measure ν_m on $\mathcal{U}(d)$ by

$$\frac{d\nu_m}{d\mu_H}(U) = c_m (\text{Tr}[U \rho U^\dagger P])^{2m}, \quad (11)$$

where $c_m > 0$ is the normalization constant. Then the following statements hold:

1. The measure ν_m forms a unitary 1-design.
2. Moreover, for sufficiently large m , there exist probability measures μ_L and μ_R on $\mathcal{U}(d)$, and observables O_1, O_2 , such that

$$\begin{aligned} \text{Var}^{(\mu_L, \nu_m)}(\rho, O_1) &\gtrsim d^{-1} \|\tilde{\rho}\|_2^2 \|\tilde{O}_1\|_2^2, \\ \text{Var}^{(\nu_m, \mu_R)}(\rho, O_2) &\gtrsim d^{-1} \|\tilde{\rho}\|_2^2 \|\tilde{O}_2\|_2^2. \end{aligned} \quad (12)$$

This result indicates that when the probability measure of one side of the circuit at a parameter forms a unitary 1-design, its total gradient variance can be at most on the order of d^{-1} , thereby exhibiting superior trainability compared to unitary 2-designs, which are bounded by the d^{-2} scaling. The proofs of Theorem 1 and Proposition 2 are provided in the Supplementary Materials, Sec. IV.

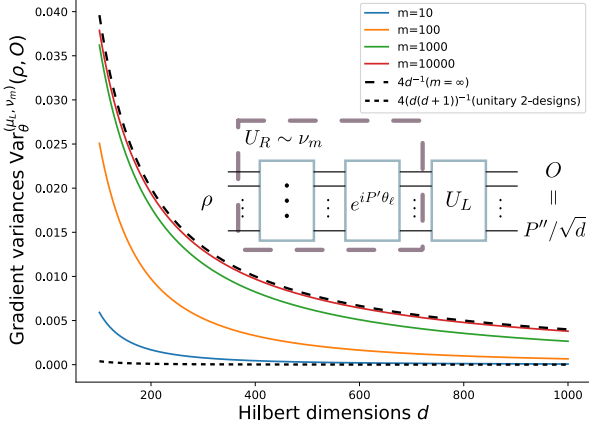


FIG. 1: The gradient variances of the circuit when adopting unitary 1-designs and unitary 2-designs as functions of the system size d . The inset shows the circuit structure used in this example. The long-dashed line represents the upper bound $4d^{-1}$ for unitary 1-designs in this circuit and the colored solid lines correspond to unitary 1-designs with μ_R taking different ν_m . The gradient variance gradually approaches the upper bound as m increases. In contrast, when the circuit adopts a unitary 2-design, the gradient variance is fixed at $4(d(d+1))^{-1}$ (short-dashed line).

Furthermore, we demonstrate the attainability of the d^{-1} scaling of the gradient variance at a given parameter θ_ℓ under a specific unitary 1-design setting. Consider the rotation gate at position θ_ℓ to be $\exp(-iP'\theta_\ell)$ and the observable $O = P''/\sqrt{d}$, where P' and P'' are two noncommuting Pauli words and U_L is an operator commuting with P'' . When μ_R forms a unitary 2-design, the variance $\text{Var}^{(\mu_L, \mu_R)}(\rho, O)$ can be computed as $4(d(d+1))^{-1}$, as indicated by the short-dashed line in Figure 1. When μ_R is taken as the unitary 1-design satisfying Eq. (11) with $P = [P', P'']/2i$, the variance $\text{Var}^{(\mu_L, \mu_R)}(\rho, O)$ is bounded by the d^{-1} scaling and, as m increases (colored solid lines in Figure 1), approaches the upper bound $4d^{-1}$, marked by the long-dashed line.

The Improvement Potential.— Beyond characterizing the trainability of a given PQC, it is also of interest to understand how to enhance trainability within a fixed circuit architecture. So far, several methods have been developed to enhance trainability, such as the substitution by shallow circuits [35, 49], tailored problem-specific ansatz [50, 51], and optimized initialization strategies [52]. However, these approaches typically rely on global or heuristic modifications and provide limited quantitative guidance within a fixed circuit structure. Here, we propose a formula to characterize the maximum improvement

in trainability through adjacent insertion of parametric oracles. Specifically, we insert a parametric oracle S_p associated with the probability measure μ on the right side of U_R . The probability measure for the combination of U_L and S_p is the convolution $\mu_L * \mu$. The schematic diagram of the process is shown in Figure 2.

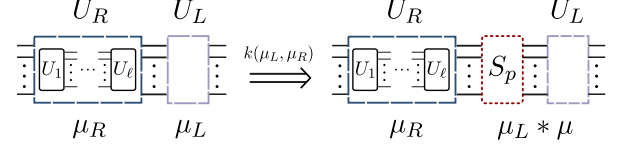


FIG. 2: The process of inserting a parametric oracle at a given position to potentially enhance trainability. Here, S_p denotes an arbitrary parametric unitary oracle associated with probability measure μ and the improvement potential $k(\mu_L, \mu_R)$ represents the maximal improvement of the total variance during the process.

Formally, we consider the following quantity

$$k(\mu_L, \mu_R) := \sup_{\mu \in \mathcal{P}(\mathcal{U}(d))} \frac{\text{Var}^{(\mu_L * \mu, \mu_R)}(\rho, O)}{\text{Var}^{(\mu_L, \mu_R)}(\rho, O)}, \quad (13)$$

which characterizes the maximal relative enhancement of variance at the parameter θ_ℓ by inserting an oracle. If $k(\mu_L, \mu_R) = 1$, the PQC exhibits no improvement potential at θ_ℓ ; whereas $k(\mu_L, \mu_R) > 1$ indicates the existence of improvement potential at that parameter. We provide the following criterion to determine whether $k(\mu_L, \mu_R) > 1$ or not:

Theorem 2. For a PQC at θ_ℓ with the probability measure pair (μ_L, μ_R) , the improvement potential is

$$k(\mu_L, \mu_R) = \frac{\|h * g\|_\infty}{\|hg\|_1}. \quad (14)$$

Here g is the Radon–Nikodym derivative $d\mu_L/d\mu_H$ and h represents the variance obtained after integrating over the right unitary defined by

$$h(V) = \int d\mu_R(U_R) \text{Tr}[V\rho V^\dagger i[U_R \rho U_R^\dagger, H_\ell]]^2. \quad (15)$$

Both functions can be obtained by the pair (μ_L, μ_R) .

Moreover, we find that the quantity $\text{Var}^{(\mu_L, \mu_H)}(\rho, O)$ provides a reference threshold. This threshold can be used to assess whether improvement is possible at a given parameter. For a pair of probability measures (μ_L, μ_R) , if $\text{Var}^{(\mu_L, \mu_R)}(\rho, O) < \text{Var}^{(\mu_H, \mu_R)}(\rho, O)$, then we can prove that the circuit must possess improvement potential at θ_ℓ . In contrast, for circuits where μ_R satisfies the unitary 2-design property, the gradient variance satisfies $\text{Var}^{(\mu_L, \mu_R)}(\rho, O) = \text{Var}^{(\mu_H, \mu_R)}(\rho, O)$. In this case, such circuits exhibit no training potential, i.e., $k(\mu_L, \mu_R) = 1$. Therefore, all parameterized quantum circuits, after being optimized through the process illustrated in Figure 2, can achieve performance at a given parameter position

that is no worse than that of unitary 2-design circuits. When $\text{Var}^{(\mu_L, \mu_R)}(\rho, O) \geq \text{Var}^{(\mu_H, \mu_R)}(\rho, O)$, no definitive conclusion can be drawn from this comparison alone. In this regime, the presence or absence of improvement potential must be determined using the criterion established in Theorem 2. More detailed discussions and constructive examples are provided in the Supplementary Materials, Sec. V.

Numerical Simulation.— To demonstrate the applicability of our zonal-polynomial formalism to practical quantum circuits, we perform Monte Carlo simulations on three hardware-efficient PQC architectures and compare the results with the predictions of our polynomial formalism. Since loss concentration implies partial-derivative concentration [53], the cost variance can serve as a proxy for diagnosing the scaling behavior whenever only the asymptotic variance scaling is relevant. Accordingly, we evaluate either the gradient variance with respect to a parameter, $\text{Var}[\partial_\ell C]$, or the cost variance $\text{Var}[C]$ for simplicity.

The three circuit architectures consist of alternating layers of single-qubit rotations and fixed entangling blocks, as illustrated in Fig. 3. In all simulations, the variational parameters are initialized independently and uniformly on $[0, 4\pi)$. In the first setting, each variational layer contains independent R_Y and R_Z rotations on all qubits followed by a fixed entangling unitary. Our theory predicts that the gradient variance associated with any parameter in the final rotation block of each layer is bounded by $O(d^{-1})$. This prediction is in good agreement with the Monte Carlo results shown in Fig. 3d. In the second and third settings, each variational layer contains independent R_X rotations on all qubits followed by a CNOT-ring or CNOT-chain block, respectively. Let $CX_{i \rightarrow j}$ denote a CNOT gate with control qubit i and target qubit j . The CNOT chain is defined as $W_{\text{chain}} = CX_{0 \rightarrow 1} CX_{1 \rightarrow 2} \cdots CX_{n-2 \rightarrow n-1}$, while the CNOT ring is obtained by adding the periodic connection from the last qubit to the first, $W_{\text{ring}} = CX_{0 \rightarrow 1} CX_{1 \rightarrow 2} \cdots CX_{n-2 \rightarrow n-1} CX_{n-1 \rightarrow 0}$. Here the gates are applied in the order written. For these two architectures, our theory yields explicit formulas for the cost variance for arbitrary n and L . The resulting predictions agree with the Monte Carlo simulations in Figs. 3e and 3f. Details of the zonal-polynomial analysis underlying the numerical simulations are provided in the Supplementary Materials, Sec. VI.

Conclusion and discussion.— Trainability is a key metric for evaluating the performance of VQAs, as it determines whether the circuit parameters of the underlying PQCs can be efficiently updated to a desirable solution. In this Letter, we developed a measure-theoretic framework based on balanced polynomial measures to characterize the trainability of arbitrary PQCs. Through zonal polynomial decomposition, this framework yields explicit expressions for the gradient variance and proposes a level classification system for trainability. Our work breaks through the strong randomness assumptions of the existing unitary design framework, enabling the trainability analysis of highly structured, practically meaningful PQCs and thereby paving the way for circuit design on near-term quantum hardware.

Looking ahead, by integrating dynamical Lie algebra theories [37, 54, 55], we expect to establish a local theoretical framework tailored for large-scale complex circuits. This approach could reduce the computational complexity of trainability analysis for specialized quantum circuit structures. In addition, the framework is well-suited to the study of circuits with symmetries [33, 56, 57], which we leave for future work. Finally, our approach can be extended to other ansatz, such as the unitary coupled cluster ansatz [58–60] and the Hamiltonian variational ansatz [61, 62], further enhancing its applications to a wider class of experimental systems.

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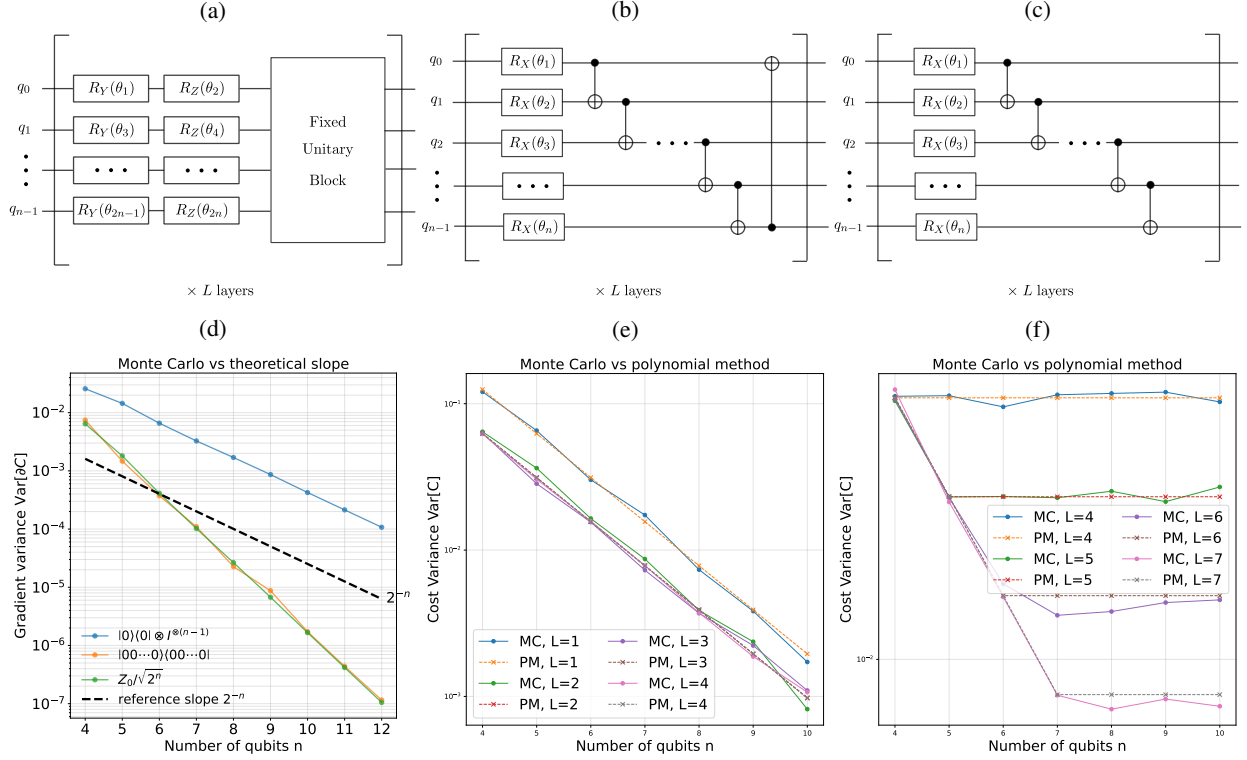


FIG. 3: Simulation setup and numerical comparison for three hardware-efficient PQCs. (a)–(c) Circuit architectures consisting of L alternating layers of single-qubit rotations and fixed entangling blocks. (d) Gradient-variance scaling for the $R_Y R_Z$ circuit. The colored solid lines show Monte Carlo estimates for different choices of observables, while the bold dashed line indicates the reference scaling $2^{-n} = d^{-1}$. The results show that the gradient variance remains bounded by $O(d^{-1})$. (e), (f) Cost-variance comparison for the R_X circuits with CNOT-chain and CNOT-ring entangling blocks. Solid lines denote Monte Carlo estimates (MC), and dashed lines denote the predictions from our polynomial method (PM). The agreement across different qubit numbers and depths demonstrates that our formalism captures the observed cost-variance scaling.

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